

Benzene-Toluene VLE: A Guided Inquiry Activity

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This guided inquiry activity is designed to accompany the interactive Benzene-Toluene Vapor-Liquid Equilibrium (VLE) simulation. Use the simulation controls to explore the thermodynamics of binary mixtures and complete the questions below.

Learning Outcomes

By the end of this activity, you should be able to:

1. Identify saturation conditions (bubble point and dew point) on phase diagrams.
2. Apply the Gibbs Phase Rule ($F = C - P + 2$) to single-phase and two-phase systems.
3. Predict the effect of changing temperature (T), pressure (P), and overall composition (z_1) on phase distributions (V/F) and equilibrium compositions (x_1, y_1).
4. Explain the fundamental operating principles of a successive multi-stage flash (distillation) process.

Key Thermodynamic Frameworks

- **Raoult's Law (Ideal Solution):** Assumes that the vapor pressure of component i in a mixture is proportional to its liquid mole fraction (x_i) and its pure saturation vapor pressure (P_i^{sat}).

$$P_i = x_i P_i^{\text{sat}} \quad \text{and} \quad P_{\text{total}} = \sum_i P_i$$

- **Bubble Point:** The temperature (at constant P) or pressure (at constant T) where the first microscopic bubble of vapor forms from an entirely liquid phase.
 - **Dew Point:** The temperature (at constant P) or pressure (at constant T) where the first microscopic droplet of liquid condenses from an entirely vapor phase.
 - **Gibbs Phase Rule:** $F = C - P + 2$, where F is degrees of freedom, C is components, and P is the number of coexisting phases.
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Part 1: Getting Familiar with the Simulation

Set the simulation to the following initial state:

- **Temperature (T):** 90.0 °C
- **Pressure (P):** 1.00 atm
- **Overall Benzene Mole Fraction (z_1):** 0.50

1. Observe the Interface:

- What is the reported "Phase State"? _____
- Record the current values for:
 - Vapor Fraction (V/F): _____
 - Liquid Benzene Mole Fraction (x_1): _____
 - Vapor Benzene Mole Fraction (y_1): _____

2. Analyze the Schematic & Diagrams:

- Look at the Pxy and Txy diagrams. Locate the **green dot**. This represents the overall system state. Is it inside the subcooled liquid, two-phase, or superheated vapor region?

- Which component (Benzene or Toluene) is more volatile? Justify your answer using the calculated values of x_1 and y_1 at equilibrium.

Part 2: Determining Saturation Conditions

2a: Bubble and Dew Point at Constant Temperature

Keep $T = 90.0\text{ }^{\circ}\text{C}$ and $z_1 = 0.50$. Set the pressure slider to $P = 2.00\text{ atm}$.

1. What is the Phase State? _____
2. Slowly **decrease the pressure**. Watch the " V/F " readout carefully.
3. At what exact pressure does the first bubble of vapor appear ($V/F > 0$)? This is the bubble-point pressure (P_{bub}).
 $P_{\text{bub}} = \text{_____ atm}$
4. Continue to **decrease the pressure**. At what exact pressure does the final drop of liquid vaporize ($V/F = 1.0$)? This is the dew-point pressure (P_{dew}).
 $P_{\text{dew}} = \text{_____ atm}$
5. Do these values correspond to the intersections of the $z_1 = 0.50$ line with the blue (bubble) and red (dew) curves on the P_{xy} diagram? _____

2b: Bubble and Dew Point at Constant Pressure

Keep $P = 1.00\text{ atm}$ and $z_1 = 0.50$. Set the temperature slider to $T = 80.0\text{ }^{\circ}\text{C}$.

1. What is the initial Phase State? _____
2. Slowly **increase the temperature**.
3. Find the bubble-point temperature (T_{bub}) where V/F starts to exceed 0.
 $T_{\text{bub}} = \text{_____ }^{\circ}\text{C}$
4. Find the dew-point temperature (T_{dew}) where V/F reaches exactly 1.0.
 $T_{\text{dew}} = \text{_____ }^{\circ}\text{C}$

Part 3: Testing Gibbs Phase Rule ($F = C - P + 2$)

For a binary mixture of Benzene and Toluene, the number of components (C) is 2.

1. **Single-Phase Region (Liquid):** Set $T = 80.0\text{ }^{\circ}\text{C}$, $P = 1.50\text{ atm}$, $z_1 = 0.50$.
 - Number of phases present (P): _____
 - Calculate degrees of freedom (F): _____
 - *Interpretation:* Change T and P by small increments. Does the system remain in the liquid state? Explain why.

2. **Two-Phase Region (VLE):** Set $T = 90.0\text{ }^{\circ}\text{C}$, $P = 1.00\text{ atm}$, $z_1 = 0.50$.
 - Number of phases present (P): _____
 - Calculate degrees of freedom (F): _____
 - *Interpretation:* With T and P kept completely constant, slide the overall composition (z_1) back and forth.

- Do the individual phase compositions (x_1, y_1) change? _____
- What actually changes in the piston container as you move z_1 ?

Part 4: Thermodynamic Sensitivities

1. **Thermal Effect:** Set $T = 90.0\text{ }^{\circ}\text{C}$, $P = 1.00\text{ atm}$, $z_1 = 0.50$. Increase T to $95.0\text{ }^{\circ}\text{C}$.
 - How did the vapor fraction (V/F) respond? _____
 - Did the remaining liquid phase composition (x_1) become richer in Benzene or Toluene? Why?

2. **Barometric Effect:** Reset to $T = 90.0\text{ }^{\circ}\text{C}$, $P = 1.00\text{ atm}$, $z_1 = 0.50$. Decrease P to 0.90 atm .
 - How did the vapor fraction (V/F) respond? _____
 - Observe the physical piston schematic. Explain how the pressure reduction correlates with the change in piston height and phase volume.

Part 5: Multi-Stage Flash Operations (Distillation)

A flash vessel splits a pressurized mixture into liquid and vapor streams at equilibrium. Let's trace how successive flash operations can separate our mixture.

1. **Stage 1 Flash:**
 - Feed: $z_1 = 0.50$. Conditions: $T = 95.0\text{ }^{\circ}\text{C}$, $P = 1.00\text{ atm}$.
 - Record the compositions of the exiting streams:
 - Vapor Stream (y_1): _____
 - Liquid Stream (x_1): _____
2. **Stage 2 Flash (Condensing the overhead vapor):**
 - Let's take the vapor stream leaving Stage 1 (y_1) and feed it as the new feed ($z_1^{\text{new}} = y_1$) into a second flash tank.
 - Adjust your z_1 slider to equal the y_1 value from Stage 1.
 - Cool this tank down to partial-condensation conditions: $T = 88.0\text{ }^{\circ}\text{C}$, $P = 1.00\text{ atm}$.
 - Record the compositions of the new exiting streams:
 - Stage 2 Vapor Stream (y_2): _____
 - Stage 2 Liquid Stream (x_2): _____
3. **Critical Analysis:**
 - Compare the initial feed composition ($z_1 = 0.50$) to the Stage 2 overhead vapor composition (y_2) and the Stage 1 liquid residue (x_1).
 - Has separation occurred? Describe how running this process across 10 or 20 successive stages relates to industrial distillation columns.

